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Project Proposal: LLM Model Compression in Numerical Linear Algebra Context

Shiwei shared an excellent [survey](#) on LLM model compression, relevant for cases where large model sizes and the computational demands are for target hardware. We aim to investigate how these compression techniques can reduce computational costs while maintaining comparable baseline performance through techniques such as Quantization, Pruning, Knowledge Distillation, and Low Rank Approximation. Mason has an NVIDIA 3060 GPU and wishes to take existing algorithms, discuss them in the context of flops, communication costs, derive error bounds, as well as benchmark them using standard metrics like perplexity scores and standard benchmark suites. The goal is to use the Llama 3.1 8B model and create compressed models for them using the 4 methods above. It might also be possible to perform [the benchmarking with distributed training](#) with a 3060 and two macbooks for an interesting training environment and also further parallel algorithm analysis.

1. Quantization: Quantization can be using 8bit floats for training, which has been discussed frequently in class as being an avenue of interest for the industry.
2. Pruning: Pruning can be used to make the matrices more sparse, and we can apply theory about random matrices and sparse matrices in class to analyze this class of compression.
3. Knowledge Distillation: This could be explored further.
4. Low Rank Approximation: Based on the seminal 2021 paper, [LORA: LOW-RANK ADAPTATION OF LARGE LANGUAGE MODELS](#), Other papers include

[LQ-LORA: LOW-RANK PLUS QUANTIZED MATRIX DECOMPOSITION FOR EFFICIENT LANGUAGE MODEL FINETUNING](#)
[LORC:Low-Rank Compression for LLMs KV Cache with a Progressive Compression Strategy](#)

We can also analyze the sensitivity of the k-rank approximation on the benchmark results.

All of these methods will be run in the same test environment and can be directly compared and insights about approximations of the full model can be drawn.